Efficient EEG-Based Motor Imagery Classification via Wavelet Genetic Optimization and Machine Learning

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***Abstract*—This paper presents a Brain–computer interfaces (BCI) system which is based on Berlin BCI Competition IV dataset built using Electroencephalography (EEG) recordings of healthy participants when performing motor imagery (MI) tasks using hand and foot movements. We applied an innovative hybrid framework which extracts time-frequency information using the Discrete Wavelet Transform (DWT) and selects features using wavelet genetic optimization (GA) reducing the data dimensions and maximizing the efficiency. This innovative combination of DWT and genetic optimization for feature selection resulted in dimensionality reduction by 2.06 times while achieving an accuracy of 89.76% with Support Vector Machine (SVM) classifier, surpassing the performance of related hybrid approaches. A comparison with weighted k-NN and wide neural networks is also provided that demonstrates robustness and efficiency of the proposed method. This makes it an effective method for classifying MI tasks and suitable for low-latency and resource-constrained real-world BCI applications.**

Keywords — *machine learning, bci, wavelet genetic algorithm, classification, discrete wavelet transform (dwt), motor imagery tasks.*

I. INTRODUCTION

Brain–computer interfaces (BCIs) provide an effective method for individuals with motor disabilities to interact with the external world. This allows a direct communication between the brain and external devices without the involvement of traditional neuromuscular routes [1]. EEG is attractive because of its noninvasive approach, portability, and cost [2], [3]. Recent technological advancements have enabled the development of robust and inexpensive BCI systems. This has resulted in an increased use of such systems in numerous applications such as neurorehabilitation, assistive technologies, and human–machine interaction. Although EEG signals offer high temporal resolution suitable for extracting useful information, they are challenging to analyze owing to the noisy, non-stationary, and high-dimensional nature of the data. This problem can be solved using advanced signal processing methods such as wavelet transform and genetic algorithms. The former is very useful in EEG signal processing because of its ability to analyze non-stationary signals in both time and frequency domains. The latter, inspired by natural evolution, are widely used for optimization problems such as feature selection. A combination of these techniques known as hybrid signal processing and optimization techniques have also shown good potential for addressing the non-stationarity and high dimensionality of EEG signals. However, many existing approaches suffer from excessive feature dimensionality or limited generalization. This motivates the development of compact and efficient feature-selection strategies customized for MI classification.

In thus study, we propose a framework that integrates wavelet decomposition for feature extraction, genetic algorithms for optimal feature selection, and machine learning classifiers for the accurate recognition of motor imagery tasks improving the reliability of EEG-based BCI. The integration of wavelets transforms, and GA reliably interprets complex biological signals. The machine learning models, such as SVM, decision trees, and neural networks, classify motor imagery and other cognitive tasks with better accuracy, thus forming the backbone of many modern BCI systems. This improves the classification accuracy of the BCI system and minimizes the computational complexity as well. We also describe a scalable roadmap and interpretable technique appropriate for real-time assistive technology applications by evaluating the impact of each component of the proposed pipeline.

The rest of this paper is structured as follows: related work is discussed in Section II, while methodology is elaborated in Section III. Section IV details the simulation results, and discussion. Section V concludes the paper and proposes directions for future research.

II. RELATED WORK

BCI technology with new paradigms, such as virtual and augmented reality, has become an active research area recently. Combining BCI with immersive environments creates new opportunities for virtual offices, civilizations, and lifestyles, as noted in [4]. The potential of BCI to support motor and cognitive recovery in the context of rehabilitation was highlighted in [5]. The development of BCI systems from early demonstrations to present uses and potential future developments is traced chronologically in [6]. Currently, BCIs are used in many different areas, such as controlling robotic prostheses, computers, wheelchairs, and cursors [11], [12], [13], and they have the prospect to enhance the freedom and standard of living of people with impairments.

A. Classical EEG-Based Motor Imagery Classification

Because of its price and portability, EEG remains the mainstay of noninvasive BCI systems. Since Jacques Vidal's 1973 demonstration of EEG-based cursor control [6] and Farwell and Donchin's 1988 P300 speller [8,9], significant strides have been made in the field. Understanding cognitive activity in interactive tasks has also been aided by other modalities, including fMRI [10]. While wearable EEG sensors now enable more natural and inconspicuous BCI solutions, practical EEG applications include intent-based cursor movement [12] and the online control of assistive robotic limbs [11]. Despite these advantages, current EEG-based BCI systems face challenges due to the non-stationary nature of EEG signals and their susceptibility to noise [7]. The model generalization is affected due to inter-subject variability. It

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extends setup time, and calibration reduces usability. Finally, real-time deployment of high-performing algorithms is limited as they are computationally intensive, necessitating the hybrid feature-extraction and optimization methods for enhancing accuracy and responsiveness of the system

B. *Wavelet-Based Feature Extraction Approaches*

Many existing works used signal-processing techniques to extract discriminative EEG features for motor imagery classification, including classical approaches such as Common Spatial Patterns and Fast Fourier Transform for identifying dominant frequency components. However, these methods often fail to capture transient temporal features. The DWT counters this problem by decomposing signals into time–frequency sub-bands. This provides better localization and improves performance for non-stationary EEG signals. For example, Subasi and Qaisar (2021) achieved 87% accuracy using a DWT–SVM ensemble model, confirming wavelet-based methods' effectiveness for MI-based BCIs.

C. *Hybrid Optimization and Machine Learning Techniques*

Machine learning classifiers remain central to EEG-based MI recognition. Traditional models such as SVM and k-NN have achieved strong results on small datasets owing to their interpretability and simplicity. Deep learning models, including CNNs and RNNs, can automatically extract spatio-temporal features but often require large, labeled datasets and powerful hardware, limiting their use in low-resource or portable BCI systems. To overcome these challenges, hybrid pipelines combining lightweight classifiers with optimized feature sets have emerged as a reliable direction for efficient and generalizable BCI systems.

In this study, an effective, noninvasive framework for motor imagery (MI) task classification using EEG data is proposed. This hybrid approach combines a Genetic Algorithm, in contrast to ant colony optimization [13], to choose the most pertinent characteristics with the DWT to extract time-frequency features. Three well-known classifiers such as WNN, SVM, and Weighted k-NN were used to assess the improved features. To guarantee reliability, the performance was thoroughly evaluated on a benchmark EEG MI dataset using various assessment criteria and cross-validation. The goal of this strategy is to support the creation of reliable and expandable BCI systems that are appropriate for assistive and rehabilitative applications.

III. METHODOLOGY

The pipeline designed for the BCI system to classify motor imagery tasks is shown in Fig. 1. The proposed framework explicitly separates feature extraction, optimization, and classification stages, enabling modular analysis and scalability. EEG dataset is acquired in the first step; noise is filtered and then normalized to improve the quality of the signal. We then use DWT to extract features by decomposing the EEG signals into multiple frequency sub-bands (e.g., SB0, SB1, ..., SBn) followed by the extraction of relevant statistical and energy-based features from the subbands to construct initial feature set. GA is applied to select the most informative features to improve the classification efficiency and reduce the dimensionality resulting in optimal feature set. It was used to train and evaluate three different classifiers such as SVM, K-NN, and WNN where each classifier predicts the category corresponding to a specific MI task. This integrated approach promises a robust, scalable, and high-performance BCI system capable of translating EEG signals into actionable outputs for assistive applications.

D. *Dataset*

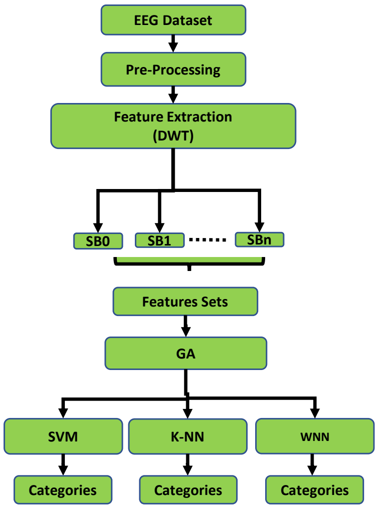

Fig. 1: Flowchart illustrating the EEG signal classification framework

This study used the BCI Competition IVa dataset [14] and includes EEG recordings from multiple subjects performing MI tasks. This dataset is widely adopted as a benchmark in MI classification research, enabling direct comparison with existing wavelet- and optimization-based approaches. Each participant performed 140 trials for two MI tasks (hand and foot movements), resulting in 280 trials per participant. Each trial lasted for 3.5 seconds which was recorded using 118 EEG channels at a 1000 Hz sampling rate, filtered between 0.5–100 Hz. The dataset provides separate training and test data for each participant to carry out systematic evaluation. Although all the participants constituting the dataset were healthy with no impairment, but it may serve as a reliable baseline for algorithmic validation before use in clinical settings.

E. *Pre-processing*

We performed pre-processing on the signal using a Discrete-Time Finite Impulse Response (FIR) filter, as shown in (1), generating an output sequence which is the sum of weighted input values [15].

$$y[n]=\sum_{i=0}^{N-1} b_i.x[n-i] \tag{1}$$

where $x[n]$ is the input signal, $y[n\,]$ is the output signal, N is the filter order, b_i is the value of the impulse response at the i^{th} instant. To suppress the irrelevant information, we applied a window function that isolates specific segments of the signal and sets all values outside the desired to zero [16] providing only relevant information for further processing.

F. *Discrete Wavelet Transform (DWT)*

The windowed signal is further decomposed into sub-bands using Wavelet Transform (WT). This mathematical transformation is defined in (2) as follows [17].

$$W_x^{\psi}(u,s)=\frac{1}{\sqrt{s}}\int_{-\infty}^{+\infty} x(t)\psi^*\left(\frac{(t-u)}{s}\right)dt \tag{2}$$

where s represents the dilation parameter, $\psi(t)$ is the mother wavelet function, and u is the translation parameter.

We used DWT to recognize patterns from EEG signals by offering a thorough time–frequency representation of the data. We used both high-pass $h[n]$ and low-pass $g[n]$ digital filters. The low-frequency components obtained are then decomposed to provide a multi-resolution representation of the signals. For acquiring coarse and fine details of the EEG signals, Daubechies wavelets were used which produces the approximation coefficients a_m^i and detail coefficients d_m^i. The former reflects the general trend of the signal, and latter captures the high-frequency characteristics. These coefficients are computed using (3) and (4), which provide the basis for effective feature extraction in the classification process.

$$a_m^i = \sum_{k=1}^{K_g^i} xf_n^i . g_{2n-k} \tag{3}$$

$$d_m^i = \sum_{k=1}^{K_g^i} xf_n^i . h_{2n-k} \tag{4}$$

Where m represents the level of decomposition.

We analyzed each subband using various feature extraction techniques, including statistical measures (skewness, standard deviation, and mean absolute value) and time-domain features (root mean square, absolute peak-to-peak difference, and zero-crossing rate). The computation of these features is detailed in (5), (6), and (7).

$$RMS = \sqrt{\frac{1}{M}\sum_{k=1}^{M} IA(i,k)^2 \forall f_i} \tag{5}$$

Where, M shows the number of samples of the windowed signal and f_i is the frequency of the i^{th} component.

$$APPD = |x_{max} - |x_{min}|| \tag{6}$$

In the above equation, x_{max} and x_{min} indicate the maxima and minima of the segmented signal, respectively.

$$ZC = \begin{cases} 1, & \text{if, } (x_k>0 \text{ AND } x_{k+1}<0 \\ & \text{OR } (x_k<0 \text{ AND } x_{k+1}>0 \\ 0, & \text{otherwise} \end{cases} \tag{7}$$

High dimensional features pose many challenges to model performance and generalization. This problem is overcome by reducing dimensionality of the dataset by using feature selection method. This method removes redundant or unnecessary features and creates a more effective model which has reduced computational complexity and also increases the generalization and learning performance, lowers class ambiguity and improves classification accuracy. It is crucial to ensure that the condensed feature set preserves the most important information from the original dataset, retaining as much data variability as feasible. This helps in preserving the integrity and efficacy of the model.

G. Genetic Algorithm (GA)

GA is a search and optimization algorithm often used to solve challenges involving feature selection and artificial intelligence with large and complex search space. GA-based feature selection process is highlighted in Fig. 2. The candidate feature subsets are refined through fitness evaluation and evolutionary operators. The starting point is population which is a collection of potential solutions represented as chromosomes stored as parameter vectors or binary strings. The fitness function measures each chromosome's ability to solve a given problem. The algorithm selects the best individuals to pass on their genetic material to the next generation based on these fitness values. When creating a new child, the crossover operator joins the segments of the two-parent chromosome. This process enhances diversity and allows the searching of unexplored regions of the solution space. The selection, crossover, and mutation processes are iteratively applied over multiple generations until a predefined termination condition is met. The predefined condition can be reaching the maximum number of generations or achieving a desired fitness level.

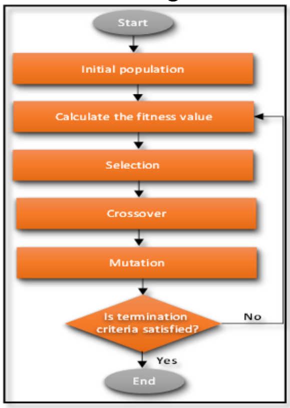

Fig. 2: Flowchart of the Genetic Algorithm process

The GA was implemented to identify an optimal subset of discriminative EEG features from the original feature space. The algorithm employed a population size of 50, mutation probability of 0.05, and crossover probability of 0.8, operating for 100 generations or until the average population fitness plateaued for five consecutive iterations. The fitness function was defined as the mean classification accuracy obtained via a 10-fold cross-validation on the SVM classifier. Binary chromosome encoding was used, where "1" indicated the inclusion of a feature and "0" its exclusion. Elitism was applied to preserve the top 10 % of individuals in each generation. This configuration was selected empirically to balance exploration and convergence speed.

H. Classification

The experiments conducted in this study used a 10-fold cross-validation [19] for a balanced and unbiased evaluation. The data were divided into ten equal subsets randomly. The nine-fold data were used for training and one for testing in each iteration. This process was repeated ten times so that every subset served as a test set at least once, contributing to the final performance metrics computed as the average across all folds to provide robust and statistically meaningful results. A polynomial kernel of degree 3 complexity parameter of 50 was employed for SVM and k-NN utilized Euclidean distance with k = 11 with class weighting proportional to the inverse of the neighbor distance. The WNN had two fully connected hidden layers (128 × 64 neurons) with ReLU activation and an output Softmax layer. We used Adam optimizer in the training with a learning rate of 0.001 for 200 epochs and simulations were performed using MATLAB R2021a with fixed random seeds.

- *Support Vector Machine*

The SVM classifier can handle both linear and nonlinear classification tasks by constructing optimal hyperplanes that maximize the margin between different classes in the feature space. If perfect separation is not feasible, then soft margin SVMs allow for some misclassifications to increase generalization and model flexibility. More generally, the support vector classifier operates as a hyperplane tailored to the dimensionality of the input data: a point in one dimension, a line in two dimensions, a plane in three dimensions, and so on. The SVM was employed with a polynomial kernel, and the complexity parameter value of 50 was found to be the most suitable. The mathematical formulation of this separating hyperplane is given by (8).

$$h(x)=\text{sign}(\omega.x^T+b) \quad (8)$$

where x is a sample vector $x=[x_1,x_2 \dots x_p]$ with p attributes, $\omega=[\omega_1,\omega_2 \dots \omega_p]$ is the weight vector, and b is a scalar bias.

- *k-Nearest Neighbors*

The k-NN algorithm is a simple yet efficient categorization method which operates on the principle that alike data points are inclined to fit into the same category. To classify a new instance, k-NN computes the distance among the new point and all points in the training set, recognizes the k closest neighbors, and assigns the class label that is most frequent among them. For example, if k=1, the algorithm designates the class of the single nearest neighbor. When $k=15$, the class is decided by a majority vote among the 15 nearest neighbors [21]. In this study, k-NN with the Euclidean distance with value of k=11 was used.

Letting v_j represent the sample and $< v_j, l_j >$ denote a tuple of a training sample and its label, $l_j \epsilon [1, C]$ where C is the number of classes. Mathematically, given a test sample z, the nearest neighbor j can be determined using the formula in (9):

$$\text{argmin dist}\left(d_j,t\right)\forall j=1\ldots N \quad (9)$$

- *Wide Neural Network (WNN)*

WNNs are machine learning models with a large number of neurons in their hidden layers comprising several interconnected nodes arranged into an input layer, one or more hidden layers, and an output layer. The input layer receives the data and processes them through the hidden layer using nonlinear functions such as Softplus, ReLU, or Sigmoid. The introduction of nonlinearity functions helps the network identify complex links and patterns in the dataset. The hidden layers change the input data to clean them and help the model extract important features. The output layer performs the final categorization step. In this study, a WNN with two hidden layers was used for the classification process.

IV. RESULTS AND DISCUSSION

We use many different performance metrics to evaluate the performance of the system and provide necessary insights for making optimal choices. The formula of these evaluation metrics are provided through (10-14):

$$\text{Accuracy}=\frac{\text{TP+TN}}{\text{TP+TN+FP+FN}} \quad (10)$$

$$\text{Precision}=\frac{\text{TP}}{\text{TP+FP}} \quad (11)$$

$$\text{Recall}=\frac{\text{TP}}{\text{TP+FN}} \quad (12)$$

$$\text{Specificity}=\frac{\text{TN}}{\text{TN+FP}} \quad (13)$$

$$\text{F-measure}=\frac{\text{2.TP}}{\text{2.TP+FP+FN}} \quad (14)$$

The AUC visually represents the categorization capabilities of the classifier [17]. The True Positive Rate (TPR) and the False Positive Rate (FPR) are defined using (15) and (16), respectively.

$$\text{TPR}=\frac{\text{TP}}{\text{TP+FN}} \quad (15)$$

$$\text{FPR}=\frac{\text{FP}}{\text{FP+TN}} \quad (16)$$

The efficacy of the proposed hybrid framework was assessed using a real EEG-based MI task dataset. It contains 840 instances evenly distributed between two classes creating an effective balance between the richness of features and computational efficiency. Each EEG segment was preprocessed to remove noise, followed by decomposition using DWT. This resulted in 97 extracted features per instance which were used to 47 after applying GA for feature selection. As result, dimensionality was reduced by almost 2.064 times significantly lowering the computational overhead, latency, and demands associated with data storage and classification. The reduced feature set was fed into the classification stage; with three classifiers whose performance was assessed for automatic MI task classification. The cross-validation technique was employed to counter the lack of access to well-labelled datasets.

The performance results of Gaussian SVM classifier are given in Table I. The classifier presents the classifying EEG-based motor imagery (MI) tasks. The classifier achieved an accuracy of 89.762% for the hand (C_1) and foot (C_2) classes. It indicates that the system consistently classifies MI tasks using EEG signals. The precision score for the hand and foot classes was 100% and 83.004%, respectively suggesting some false positives in latter. The recall value for the Foot class was 100% indicating that all Foot class instances were correctly identified. The Hand class, on the other hand, had a recall value of 79.524% indicating that some instances were not identified correctly. This contrast is also reflected in specificity values which are 100% and 79.524% for the hand and foot classes, respectively indicating that negative cases are correctly identified. Both hand and foot classes achieved strong F1-scores of 88.594% and 90.713%, respectively, whereas combined average was 89.654%. The AUC for both classes was same, 97.00%, showing an excellent capability to distinguish between MI tasks.

TABLE I. EVALUATION MEASURES FOR THE GAUSSIAN SVM CLASSIFIER

Performance Metric	**Hand Class (C_1)**	**Foot Class (C_2)**	**Average Performance**
Accuracy	89.762%	89.762%	89.762%
Precision	100.00%	83.004%	91.502%
Recall	79.524%	100.00%	89.762%
Specificity	100.00%	79.524%	89.762%
F1-score	88.594%	90.713%	89.654%
AUC	97.00%	97.00%	97.00%

The performance results weighted k-NN of classifier are given in Table II. The classifier achieved an accuracy of

86.190% for both classes when identifing hand and foot MI tasks showing balanced performance across different classes. The precision scores of 86.019% and 86.364% for the Hand and Foot classes indicate low rate of false positives and consistent accuracy for both classes when making predictions. Similarly, recall values of 86.429% and 85.952% indicate that most of the actual instances were identified correctly. The specificity scores of 85.952% and 86.429% for the hand and foot demonstrate high reliability in correctly detecting negative cases across both classes. An average value of 86.190% for F1-scores demonstrate balanced trade-off between precision and recall. The AUC score of 93.00% for both classes confirmed the model is robust discriminating against different MI tasks.

TABLE II. EVALUATION MEASURES FOR THE WEIGHTED K-NN CLASSIFIER

Performance Metric	Hand Class (C_1)	Foot Class (C_2)	Average Performance
Accuracy	86.190%	86.190%	86.190%
Precision	86.019%	86.364%	86.191%
Recall	86.429%	85.952%	86.190%
Specificity	85.952%	86.429%	86.190%
F1-score	86.223%	86.158%	86.190%
AUC	93.00%	93.00%	93.00%

Similarly, Table III presents an evaluation of the performance of the WNN classifier achieving an accuracy of 80.357%. This indicates that system has a moderately strong ability to classify instances from both classes. The precision recorded was 79.176% and 81.638% for hand and foot classes, respectively. The average precision of 80.407% suggests a reasonable capability to minimize false positives. The recall values of 82.381% (hand) and 78.333% (foot) revealed that system has a slightly higher sensitivity in detecting hand instances. The specificity scores of 78.333% (Hand) and 82.381% (Foot) show that the classifier is slightly more effective at identifying true positives in the foot class. The F1-scores were 80.747% for the hand and 79.951% for the foot, with an average of 80.350%, reflecting a balance between the predictive strengths and weaknesses. The AUC scores of 88.00% for both classes indicate that system has a good ability to distinguish between the Hand and Foot MI tasks, but it is lower than that of the other tested classifiers. The WNN classifier exhibited reliable but lower performance than the Gaussian SVM and weighted k-NN classifiers, making it a viable candidate for EEG-based MI classification.

TABLE III. EVALUATION MEASURES FOR THE WNN CLASSIFIER

Performance Metric	Hand Class (C_1)	Foot Class (C_2)	Average Performance
Accuracy	80.357%	80.357%	80.357%
Precision	79.176%	81.638%	80.407%
Recall	82.381%	78.333%	80.357%
Specificity	78.333%	82.381%	80.357%
F1-score	80.747%	79.951%	80.350%
AUC	88.00%	88.00%	88.00%

A comparative analysis of different classifiers based on accuracy, precision, recall, specificity, F1-score, and AUC is shown in Fig. 3. The SVM classifier performed better than the other two models in all metrics. The AUC performance is particularly better edging out the weighted k-NN, indicating that it has excellent discrimination ability. It also performed better in terms of precision, recall, and F1-score. This reflects a strong balance between correctly identifying true positives and reducing false positives and false negatives. The performance of Weighted k-NN was at par with SVM in terms of precision and AUC but lags slightly in recall and F1-score. This questions its effectiveness in capturing all true-positive instances. On the other hand, the WNN classifier showed decent performance in terms of AUC but performed poorly across all other evaluation criteria when compared to other models. This indicates that models have a higher rate of misclassifications and reduced generalization capabilities. The discussion concludes that SVM is the most robust and reliable classifier among the three for the given MI EEG classification task, followed by weighted k-NN, with WNN showing the least favorable performance.

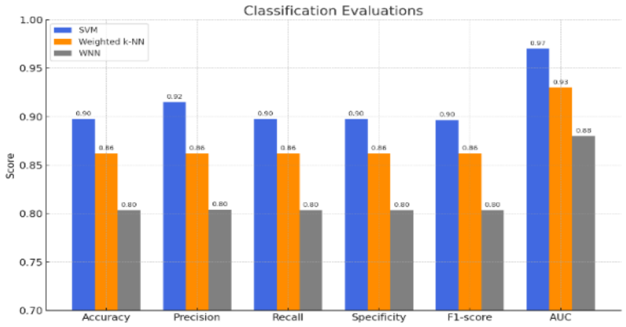

Fig. 3: The classification evaluations comparison between the three classifiers

The compression, resource utilization, and power consumption efficiency of the system can be increased tactically using event-driven and adaptive rate preprocessing and recognition approaches [15]. To improve the system's computational and classification performance, other optimizer-based feature selection and dimension reduction methods may be helpful [23]. It may be possible to examine the viability of using these tools in the suggested approach. Furthermore, the modular architecture of proposed framework can be adapted to other metaheuristics (PSO) Particle Swarm Optimization or Ant Colony Optimization (ACO) like [13], [23]. This flexibility makes the proposed DWT-GA scheme a foundation for scalable future hybrid neuro-signal processing pipelines which can be used in assistive and rehabilitative technologies.

In order to validate the significance of the observed performance differences, we conducted Wilcoxon signed-rank test between classifiers across subjects. The results indicate that the accuracy improvements achieved by the SVM classifier are statistically significant ($p < 0.05$). This is confirmed that the performance gains are not due to random variation. Also, we performed ablation analysis whose results are provided in Table IV. These results demonstrate that GA-based feature selection substantially improves classification accuracy while reducing feature dimensionality highlighting its contribution to the proposed framework.

TABLE IV. ABLATION ANALYSIS DEMONSTRATING THE IMPACT OF GA-BASED FEATURE SELECTION.

Method	Features	Accuracy	F1 Score
DWT + ML (No GA)	95	83.33%	0.829
DWT + GA + ML (Proposed)	45	89.76%	0.892

We also compared the performance of our framework with existing work in Table IV. The proposed model achieved the highest accuracy of 89.76% while reducing the feature set size by 50%, confirming its superior efficiency and generalization capability. Also, it provides a simpler and more computationally efficient pipeline when compared with ACO and other deep learning approaches making it suitable for embedded BCI systems. To contextualize the performance of the proposed framework, a comparative analysis with representative DWT-based, GA-based, and hybrid MI classification methods reported in the literature is presented in Table V.

TABLE V – COMPARISON OF PROPOSED FRAMEWORK WITH PRIOR STUDIES

Study	Dataset	Method	Acc.	Remarks
[11]	Custom 10-ch EEG	CSP + CNN	85.60%	Deep model; limited real-time feasibility
[13]	BCI Comp IVa	Wavelet + ACO + ML	88.50%	Good performance; higher computational load
[14]	BCI Comp IVa	DWT + Ensemble SVM	87.10%	High accuracy; higher feature dimensionality
[21]	EEG Motor Tasks	Weighted k-NN (metaheuristic)	86.20%	Reliable; slower training time
Proposed	BCI Comp IVa	DWT + GA + SVM	89.76%	Efficient, compact feature set (2.06× reduction)

While deep-learning approaches have shown promising results in MI classification, they typically require large-scale datasets and substantial computational resources. Also, LASSO-regularized neural networks may work but their performance is often sensitive to hyperparameter tuning and dataset size. The proposed framework offers a lightweight alternative that achieves competitive performance with significantly reduced complexity, making it suitable for real-time and embedded BCI applications with small datasets. But it suffers from limited generalization to motor-impaired users due to EEG data collected from healthy subjects. Also, the use of GA for improving feature compactness introduces computational overhead that may affect real-time deployment without further optimization.

V. CONCLUSION

This research presented a hybrid technique for the automatic classification of MI tasks using EEG signals by integrating the DWT for feature extraction, the GA for feature selection, digital filtering for denoising, and machine learning classifiers for the final classification. The main contribution integration of DWT and GA which gives a compact and high-discrimination feature set capable of capturing both temporal and spectral EEG dynamics. This not only increases accuracy but also reduces computational and memory requirements at the same time, making it suitable for low-latency BCI systems. The weighted k-NN model showed a bias toward C_2 which can be useful in applications where the accurate identification of this class is essential. In contrast, WNN, on the other hand, performed poorly and it is not feasible for deployment unless it is improved significantly. Despite promising results, it suffers from some as the of evaluation on healthy-subject datasets. Future work will focus on real-time BCI deployment, improving cross-subject generalization, and integrating hybrid deep-learning and ensemble learning methods to enhance robustness and scalability.

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